

Multiple Choice (10 marks)

- Which of the following is lower for argon than for neon?
 - Melting point
 - Boiling point
 - Polarizability
 - First ionization energy
- Calculate the equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field at 300 K.
 - 10.8×10^{-5}
 - 4.11×10^{-5}
 - 3.43×10^{-5}
 - 1.71×10^{-5}
- Calculate the Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz.
 - $Q = 1012$
 - $Q = 2012$
 - $Q = 3012$
 - $Q = 6012$
- The number of allowed energy levels for the ^{55}Mn nuclide are:
 - 3
 - 5
 - 8
 - 4
- When the following equation is balanced, which of the following is true? $\text{MnO}_4^- + \text{I}^- + \text{H}^+ \rightleftharpoons \text{Mn}_2+ + \text{IO}_3^- + \text{H}_2\text{O}$
 - The $\text{I}^- : \text{IO}_3^-$ ratio is 3:1.
 - The $\text{MnO}_4^- : \text{I}^-$ ratio is 6:5.
 - The $\text{MnO}_4^- : \text{Mn}_2+$ ratio is 3:1.
 - The $\text{H}^+ : \text{I}^-$ ratio is 2:1.

True / False (10 marks)

Answer True or False

6. True or False: Sc^{3+} has the smallest ionic radius among the ions listed due to its higher positive charge and fewer electron shells.
7. True or False: Nuclear magnetic resonance spectroscopy is a light-scattering technique that analyzes molecular structures.
8. True or False: The strongest base in liquid ammonia is the amide ion (NH_2^-).
9. True or False: Silicon is most commonly found in nature as silicon dioxide, which is an oxide.
10. True or False: Doubling the volume of a buffer solution by adding water significantly increases its pH.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to the differences in relaxation times T_1 and T_2 for a nucleus, particularly focusing on the sensitivity of T_2 to molecular motions at the Larmor frequency.
12. Discuss how the ionic strength of a solution is influenced by the concentration and dissociation of its ions, using 0.050 M AlCl_3 as an example.

End of Paper